



Cricket

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Performance Demands of Cricket

In this section, we introduce the sport of cricket and discuss the various positional demands within the game and the factors that contribute to successful performance. We also look at the relative contribution that physical performance makes toward sustaining long-term performance and winning.

Sport Description and Factors of Winning

Cricket is a bat-and-ball sport played between two teams of 11 players. Globally it is played in more than 125 countries, and it is particularly popular in the Asian subcontinent where it is the most popular sport in five countries and enjoyed by a combined population of over 1.4 billion people. Three different game formats are played at the elite level: Twenty20 (T20) is a three-hour game consisting of 20 overs per side, where an over is a set of six balls bowled by the same bowler. The One Day International format (ODI) consists of a maximum of 50 overs per side and takes ~7 h. These first two formats are also categorized as *limited overs cricket*. Last, the most traditional format is termed Test Match Cricket, which is played over 5 d, taking ~30 h. The sport is played on an oval field (minimum diameter of 137 m

[150 yd]) with a central 20.1 m (22 yd) grass pitch with wickets (a set of three wooden stumps planted in the ground with two smaller pieces of wood [bails] sitting on top of them) located at each end (figure 24.1).

Player positions are typically classified into primary skill disciplines within the game. These are described as batters, bowlers (fast, medium, spin), fielders (with a multitude of different positional names), and a wicketkeeper (see figure 24.1). The object of the game is to score runs by running between the two crease lines marked ~17.7 m (19.4 yd) apart on the pitch. To score a run a batter must generally hit the ball away from the fielders to allow enough time for the batting pair to complete one or more runs. Other methods of scoring runs include hitting the ball past the boundary line after bouncing (4 runs) or over the boundary on the full without bouncing (6 runs). The fielding side tries to end the batting team's innings early by dismissing 10 of the opposing team's batters. The most common dismissal methods are to hit the batter's wickets when bowling (bowled); to catch the ball on the full after the batter has hit it (caught out); or to field and throw the ball to hit the wickets before a batter has made his ground (within the safety of the crease zone) when attempting a run (run out). The team that accumulates the highest number of runs after both teams have either batted once (limited over formats) or twice (test format) wins.

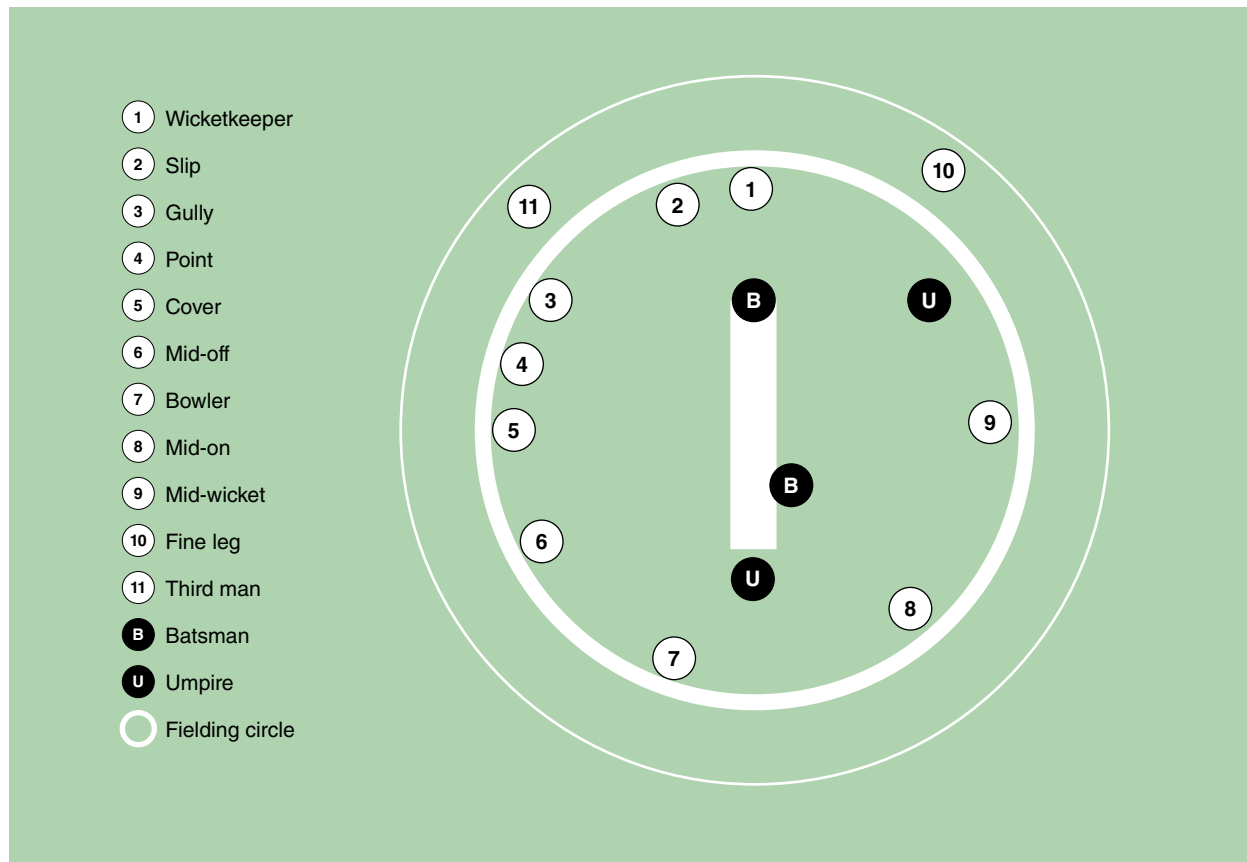


Figure 24.1 Fielding positions for a right-handed batsman.

To understand the factors that underpin team success in cricket, one must consider the tactical elements involved. Cricket is a very tactical game and revolves around how aggressively (amount of risk taking) to bat to either *set* or *chase* a total number of runs. Coaches often use performance analysis to map an innings, determining where a team should be reaching certain scores by at each stage of an inning to enhance probability of winning the game (27). Different types of hits (shots) carry different levels of risk. Key performance indicators have been used to distinguish between winning and nonwinning teams (16), and to identify times during an innings when it is better for the fielding team to take a wicket (3, 14) or for the batting team to score a higher percentage of boundaries (9). The captain (often in consultation with his bowlers) is responsible for all of the tactical decisions made on the field and has the important role of determining where the nine fielding players (excluding the bowler and wicketkeeper) are spread (placed) around the field to both maximize the chances of taking wickets and restrict the opportunities for opposing batters to score

runs. To limit players scoring runs, the bowlers on the fielding team will try to disguise the type of delivery they are bowling using movement pattern similarity (23) or by bouncing the ball in areas of the pitch from which the batter will find it hardest to score. The direction and placement of a bowler's pitch of the ball determines where on the field a batter is more likely to hit the ball (4). Captains develop knowledge of these likely hit placement locations through years of playing experience, talking to coaches and other players, consulting coaching resources, or using technology tools such as CricketPlaybook app (13).

Like so many sports, skill plays an important role in cricket success. The skill of cricket bowling is being able to execute the intended type of delivery with a high degree of accuracy and consistency, thereby allowing the captain to set his planned field placements. A bowler strives to deceive a batter into missing the ball, or playing a mistimed shot, and different types of bowlers specialize in special forms of deception. Fast bowlers may use extreme speed, while slower spin bowlers impart a large number of

sideways revolutions on the ball to induce lateral movement after the ball bounces. Bowlers can also make the ball move laterally (swing) in the air. NASA aerodynamics scientist Rabindra Mehta (8) explains that there are three types of swing (conventional, reverse, contrast), and bowlers devote many practice hours trying to maximize their swing, speed, or spin. The batting challenge is exacerbated by the limits of human reaction time with studies showing that any ball that makes an unpredictable movement less than 170 to 200 ms from the batter is physically unplayable (2, 7). Batters develop technique strategies to deal with this challenge (soft hands, closer foot placement to the ball bounce location) to reduce their likelihood of getting dismissed (28). Batters refine their technique with hours of practice to enhance their ability to judge the line and length of deliveries from different types of bowlers in different types of conditions.

Physical Performance Contributors to Winning

Coaches often describe the ultimate cricket performance as the interaction of four key factors: technical, tactical, psychological, and physical. Cricket is

described as an interceptive action sport, with batting skill heavily reliant on hand-eye coordination or reaction time. Elite batters need to move quick enough to hit a ball moving up to 160 km/h (100 mph). Cricket also requires the ability to concentrate over extended time periods and remain tactically aware of the present game situation. As an extreme example, Test batter Hanif Mohammad holds the record for the longest time spent at the crease (over 16 h, scoring 337 runs in 1958). The fact that a typical day of Test cricket consists of three sessions of 2 h each, with breaks of 40 min for lunch and 20 min for tea, Hanif would have batted for more than 2.5 d! Common to all the different game formats, and from a physical standpoint, cricket is interspersed with numerous brief explosive moments of action.

As illustrated by its central placement in figure 24.2, cricket is, above all else, a game requiring inordinate physical characteristics and mental aptitude, including the ability to concentrate intensely for very prolonged periods. However, displaying high levels of physical features cannot, on its own, fully compensate for any lack of ability of the other key factors (10). Maintaining a high level of physical fitness is increasingly important to both maximize

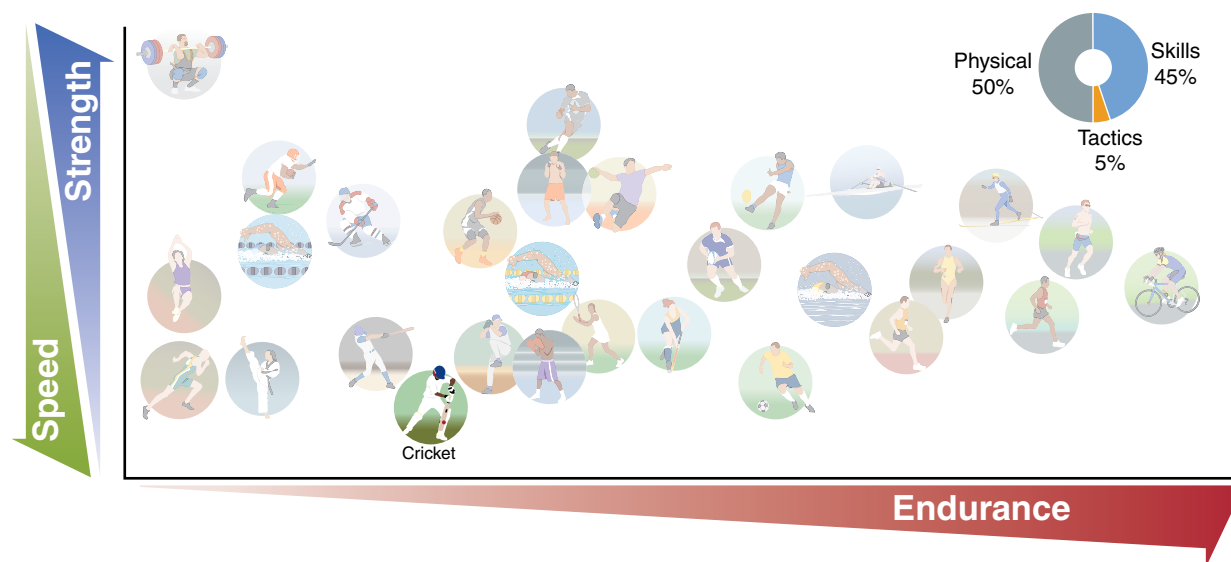


Figure 24.2 The position of the cricket player on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation, acknowledging this varies slightly between specialist positions (batters, fast bowlers, wicketkeeper). Outside of critical psychological and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills (45%), tactical awareness (5%), and physical capacities (50%) for cricket fast bowling success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

performance but also to cope with high workloads in order to minimize injury risk. Modern elite cricket players can expect to tour for up to 11 mo and play a high volume of matches ($n = 100$ d approximately) in a calendar year. The injury prevalence is higher in international players as they play for most of the year without a substantial off-season (11). Therefore, enhanced physical fitness may help a player to be physically more resilient to injuries and thereby be in a position to contribute to the team for a greater number of matches throughout the season.

Targets of Physical Performance in Cricket

Given the length of the international cricket season and touring demands, superior aerobic fitness is thought to assist players to recover and sustain elite performance. Body composition measures with tests of vertical jump; sprint split times over 5, 10, and 20 m (5, 11, 22 yd) (from a stationary start); and an aerobic field test are the most commonly used testing battery for elite cricketers. Mean data from Australian national level cricketers (20) shows running performance in the Yo-Yo intermittent recovery test (IRT) of 18.1 ± 1.8 levels (1863 ± 444 m; 2037 ± 486 yd) for males and 15.8 ± 1.2 levels (970 ± 385 m; 1061 ± 421 yd) for females, with countermovement vertical jump scores of 60 ± 9 cm (24 ± 4 in.) and 44 ± 6 cm (17 ± 2 in.), and 20 m (22 yd) sprint times of 3.06 ± 0.11 s and 3.43 ± 0.15 s (male and female international data, respectively) also reported.

Batters

All players are required to bat, so successful physical performance attributes apply to all players. Batters must judge how quickly they can complete a run compared to how quickly the fielding team can return the ball. Being a quicker sprinter between the wickets will allow for a greater number of potential run-scoring opportunities. Turning, or change of direction (COD) ability, is another important attribute and is commonly measured for each leg separately with a 5-0-5 test. Elite 5-0-5 scores of 2.27 ± 0.13 s (males, same scores for both legs); 2.49 ± 0.36 s and 2.45 ± 0.09 s (female left and right leg turn, respectively) have been reported (20). It has been postulated that the repeated eccentric muscle contractions that occur in acceleration, deceleration, and the COD required in running between the wickets is likely to encompass a large neuromuscu-

lar demand, inducing muscle damage, and hence the best way to prepare the body for these demands is to substantially increase the eccentric muscle strength of the involved musculature (28). During a 30 min innings, T20 batters have been shown to cover ~ 2.5 km (1.6 mi), while sprinting 12 ± 5 times, with a mean sprint distance of 14 ± 3 m (15 ± 3 yd), and a total sprint distance of 160 ± 80 m (175 ± 87 yd) (15). The demands of batting also require a high level of aerobic fitness in order to play for up to 6 h a day. Higher aerobic fitness also helps with better thermoregulation responses and faster adaptations when playing in hot and humid climatic conditions, which is especially important due to the protective equipment that batters choose to wear (19).

Bowlers (Fast, Medium, Spin)

Bowlers are primarily classified by their ball release speed, but we will only focus here on fast bowlers as they have the greatest physical demands. Fast bowlers generally have 15 to 30 m (16 ± 33 yd) run-ups, with those running in from the longer 30 m (33 yd) length needing to be nearly as aerobically fit as a triathlete to sustain this for 30 overs a day (28). On average, a fast bowler's run-up lasts about 4 s from when he hits his stride cruising intensity. The run-up culminates in the explosive delivery stride, when the bowler absorbs high ground-contact forces and the momentum is transferred to the 156 g (0.34 lb) ball to bowl it quickly. Global positioning system (GPS)-based time-motion analyses have shown fast bowlers during multiday matches will cover $\sim 22.6 \pm 4.0$ km (14 ± 2.5 mi) total distance in a day with 1.4 ± 0.9 km (0.9 ± 0.6 mi) at sprinting intensities (17). In comparison, ~ 13 km (8 mi) and ~ 5.5 km (3.4 mi) total distance are covered in One Day and T20 matches, respectively (17). In T20 cricket, a fast bowler can expect to sprint 42 ± 8 times, with a mean and maximal sprint distance of 17 ± 2 m (18.6 ± 2 yd) and 51 ± 20 m (56 ± 22 yd), respectively (15). Woolmer describes bowling fast "as one of the most demanding activities in the world of sport: to reach the pinnacle of the game, and prosper there, modern fast bowlers must be among the most athletic of humans" (28).

Most injury research has been undertaken on fast bowlers as this position has the greatest injury prevalence. Orchard et al. (12) found high acute workload (more than 50 overs in a match) may lead to a somewhat delayed increased risk of injury up to 3 to 4 wk after the acute overload, possibly via a mechanism of damaging immature (repair) tissue.

The most common injury is hamstring strain (seasonal incidence 8.7 injuries/100 players per season), but the most costly injury is a lumbar stress fracture, with 1.9% of players unavailable at all times owing to these injuries, representing 15% of all missed playing time (11).

Wicketkeeper

The wicketkeeper is a specialist fielding position. A wicketkeeper must catch and stop fast-moving balls, which requires anticipation, agility, speed, quick acceleration, and often diving laterally or jumping vertically. The wicketkeeper is potentially involved in every single delivery during the fielding innings and will run to the stumps in anticipation of a run-out opportunity or to catch the ball being thrown in by the fielders, with wicketkeepers taking ~41 catches from the field per 50 over innings (6). Interestingly, wicketkeepers rarely sprint long distances despite still covering a daily distance of 16.6 ± 2.1 km (10.3 ± 1.3 mi) in multiday cricket (17) and 6.4 ± 0.7 km ($4 \pm .4$ mi) in T20 (15). A vital conditioning requirement for wicketkeepers is repetitive, multiplanar, low-intensity exercise with intermittent bouts of explosive movement (6).

Fielders

All other players are required to field and can be classified as either infielders or outfielders based on how close to the central wicket area they are positioned. Outfielders will often have longer sprints, while infielders tend to have shorter sprint activity. Some specialist catching fielding positions, like slips fielders, are virtually stationary. But generally, fielders should be agile and fast to catch and stop fast-moving balls. Ideally, fielders should have a fast, strong, accurate throw that can be used to run a batter out. Fielders have been shown to cover 8.1 ± 1.3 km ($5 \pm .8$ mi) total distance in an 80 min T20 innings, with sprinting accounting for 554 ± 353 m (595 ± 386 yd) (15). A fielding player can expect to sprint 31 ± 18 times with a mean and maximal sprint distance of 17 ± 4 m (19 ± 4 yd) and 54 ± 23 m (59 ± 25 yd), respectively (15).

Key Weapons, Manipulations, and Surveillance Tools

Recall that weapons refer to the high-intensity interval training (HIIT) formats we can use to target the physiological responses of importance (chapters 4

and 5), while the surveillance tools (chapters 8 and 9) are what we use to describe and monitor the individual responses to those weapons. In this section, we present the different HIIT weapons we use in cricket, their manipulations, along with ways to monitor their effectiveness (surveillance).

Often the full spectrum of training activities undertaken across the year is not reported in cricket research. However, one study of elite academy cricketers over a 14 wk preseason preparation phase has defined 28 distinct training drills (18). These were then classified into four more general drill categories (high- and low-intensity conditioning drills, game simulations, and skill-based drills). Of the high-intensity sessions, most were running-based variations, including the square running-based drill that involved varied duration agility efforts over 20 to 60 m (22 ± 66 yd) (HIIT type 2), but Tabata-like bike sessions (HIIT type 4) were also used. One limitation of this study was the failure to account for the strength-based sessions (see chapter 6). Indeed, nonspecific gym-based HIIT drills can account for a significant proportion of training in elite programs, making up ~15% of training (24).

By far, the most common and traditional form of training for cricketers is cricket net-based training with bowlers bowling to a batter in an enclosed net. The number of bowlers sharing a net will influence the intensity of the session for each bowler. It is quite common for bowlers to bowl in a pair to mimic the time between balls in a match (20 to 30 s), and after a pair of bowlers has each bowled their six deliveries, they will often swap with another pair to replicate the between-over rest period. Batters will often utilize a bowling machine instead of bowlers in order to gain a greater volume of work or to practice against a particular type of delivery. While repeated sprints (type 5) are sometimes programmed into a net session, traditionally batters have generally tended to remain stationary. More recently, consequence (batter dismissed) shuttle sprints (of repeated 2 and 3 s = 40 to 60 m [44 ± 66 yd]) have become popular in net sessions at the international level.

HIIT Weapons

In our experience, we typically target three main HIIT types throughout the season (figure 24.3). As cricket consists of numerous brief explosive moments of action (<10 s), the most popular HIIT sessions implemented are short-interval HIIT and repeated-sprint training (RST) sessions (types 2 to 4, 5 targets).

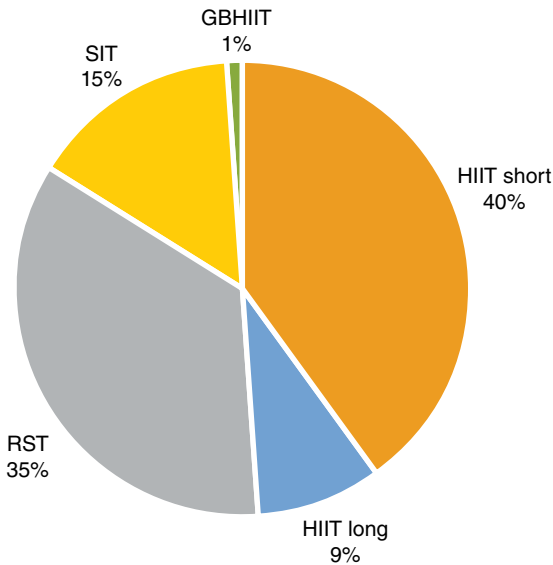


Figure 24.3 Percentage of the different HIIT weapons (formats) used throughout the annual season in elite cricket. HIIT: high-intensity interval training; RST: repeated-sprint training; SIT: sprint interval training; GBHIIT: game-based HIIT.

Additionally, prescribed year-round for their injury protective qualities, in addition to their development of speed abilities, sprint interval training (SIT) sessions (type 5 target) are becoming increasingly popular across elite teams. Finally, as building aerobic fitness is important, long-interval HIIT sessions tend to be implemented occasionally in the pre-season (type 3 and 4 targets). Game-based HIIT sessions (type 2 and 4 targets) are used infrequently and tend to be of more interest to researchers.

Short-Interval HIIT

Conditioning intervals for aerobic power and repeat effort ability have become increasingly popular with strength and conditioning coaches across elite cricket. These are programmed using a cricketer's calculated maximal aerobic speed (MAS), and commonly the prescription ranges from 10 to 30 s work bouts, using a 1:1 to 1:3 work:rest ratio (HIIT type 4). With the increasing use of GPS data across cricket, conditioning intervals are also prescribed to target specific mechanical and physiological qualities. These can be used to overload training time within specific zones according to individual need or can be used to top-up players who are not exposed to desired stimuli from match play alone.

Despite their lack of cricket specificity, other sessions employed at the elite level include Tabata

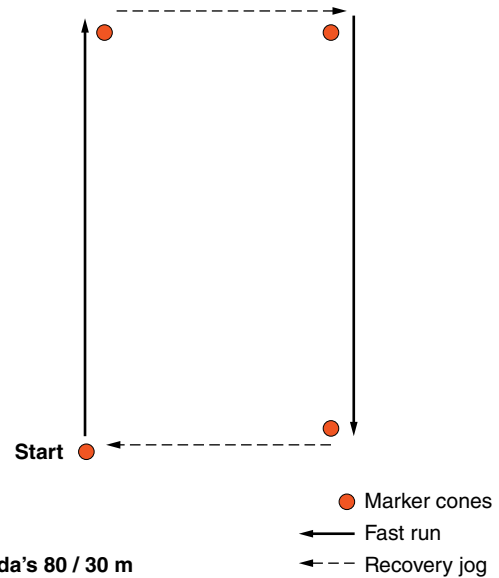


Figure 24.4 Example of an HIIT short running-based activity used for bowlers (HIIT type 1).

cycling protocols with 20 s on/10 s off (HIIT type 4). These sessions are particularly useful when trying to acclimate players before traveling to more challenging environmental conditions, and are occasionally performed inside a heat tent. Treadmill sprints using 20 s efforts with 40 s recovery periods at ~18 km/h (11 mi) are also a favorite method for providing a quick recovery-type workout (HIIT type 1) as is Chaminda's 80/30 m drill (HIIT type 1; figure 24.4).

Repeated Sprint Training

In our experience, RST is the most commonly used HIIT type in cricket. Coaches design interval sprints to simulate the run-up approach of fast bowlers, fielding drills to work on accelerations and handling skills, and running between the wickets for batters. Pyke and Davis (22) in their book *Cutting Edge Cricket* recommend that fast bowlers complete a series of 25 to 30 m (27 ± 33 yd) sprints at about 90% of top speed with 15 s recovery periods between them and sets of 6 to 8 sprints should be interspersed with 60 to 90 s breaks between sets (HIIT type 4). During the preseason, coaches will progressively increase these bowl-throughs to simulate the expected bowling loads (number of overs) that the bowler is expected to experience in a game. It is prudent for bowlers to undertake these on similar surfaces to what they will be bowling on. Often bowlers may build up their RST volumes on softer winter grounds and make the mistake of failing to adjust their RST training

volume when they travel to their preseason training camp and start training on summer-hardened foreign pitches, the progression of neuromuscular loading suddenly becoming inappropriate. In-season RST is often used for individual top-ups.

Using published cricket-specific time-motion data, Houghton et al. (5) released an app (BATEX) that simulates scoring a one-day cricket century through verbal instructions and audio beeps. The BATEX app develops fitness in batsmen by adding realistic running to net sessions, by controlling the number of runs and the pace of running through six different stages. The BATEX app has been utilized to compare batting skill performance between different competitive levels. The findings from this study showed a 12% better percentage of good bat-ball contacts were made by higher-grade compared to the lower-grade batsmen (5), which may indicate a possible relationship between fitness, concentration, and skilled performance.

Simulated batting innings have been a common training methodology employed by conditioning coaches, especially in the preseason. Armed with a stopwatch and yelling a predetermined number of runs (from 1 to 4) players are conditioned to the specific demands of accelerating and decelerating and working on the technique of quickly turning between the wickets. Progressively increasing the cumulative number of runs undertaken in a session adjusts volume. To add a competitive element, the

drill is often completed in pairs or even within a group setting. A required target time completion incentive for these run repetitions is also often provided. Using the BATEX protocol of scoring a simulated century, 17 university-level male batters were found to have heart rates of 124 ± 15 to 159 ± 14 beats/min, with measured oxygen uptakes of 29.3 ± 6.1 to 43.4 ± 6.3 mL·kg⁻¹·min⁻¹ and core temperature readings of 37.7 ± 0.3 °C (32 °F) to 38.7 ± 0.4 °C (33 °F) (21). Yet, generally practitioners will only use performance times and occasionally heart rate to gauge progress.

Numerous fielding drills can additionally be classified as RST. The intensity and recovery periods depend on the number of players involved in the drill, the throwing intensity being employed, and the execution of the involved skills. A common fielding drill is around the world, which involves players taking turns to stop an incoming ball, shy (throw accurately and quickly) at the stumps, sprint to the next cone, and back up the incoming throw (HIIT type 5; figure 24.5). Another drill we use particularly in-season during the warm-up is the Southern Cross Energizer (figure 24.6). This drill involves two players simultaneously performing a quick 5 s agility run with either one or two catches or one-handed ground pickups, and encourages quickness of movement and accuracy of fielding skill execution. Fielding high ball catches and boundary ground fielding can be manipulated via the use of both acceleration

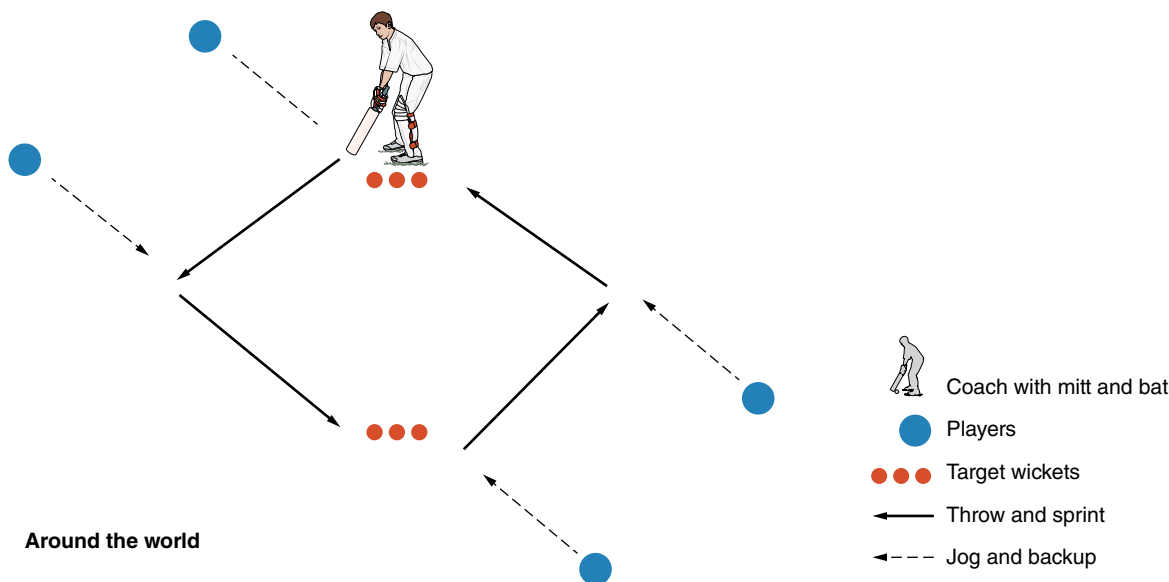


Figure 24.5 Example of an HIIT fielding activity incorporating shying at the wickets, sprinting, and rotating around the field to back up the throws (HIIT type 4).

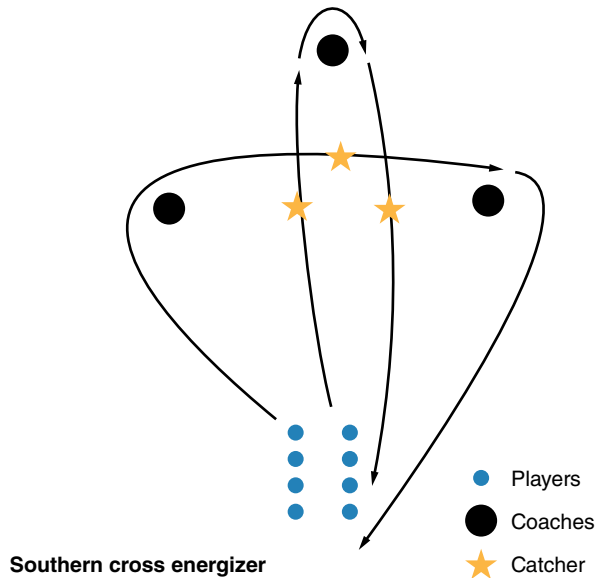


Figure 24.6 Example of a pregame HIIT activity incorporating fielding, catching, and/or ground fielding (HIIT type 4).

zones and/or strategic feeding of balls to challenge an individual's ability to make the catch or ground ball at a desired sprint speed. These can be done with specific work:rest ratios to elicit targeted physiological responses.

Long-Interval HIIT

Using the circumference of the playing oval (~500 m) (547 yd), conditioners often prescribe long-interval HIIT using full repetitions of this distance (~90 to 95 s for elites) in the preseason period, primarily because of their important neuromuscular load. The intensity is normally at a bowler's striding pace and sometimes monitored using heart rate. However, more commonly used instructions are to maintain a set RPE or to try to replicate the last repetition (recovery time of 1 to 2 min). The number of repetitions is increased up to 10 as the preseason progresses. On-field high-intensity running sessions (for example, colored cones indicating varied intensity and distances of sprints and recoveries) are also employed in the team environment during the in-season to provide a fitness top-up (HIIT type 4).

Gym-based cycling (HIIT type 1) is also commonly used to provide a cardiovascular top-up, especially for players who need to limit lower-body impact. Long-interval HIIT is also used for muscular endurance (HIIT type 4) across a variety of exercises such as weighted chin-ups and bodyweight

exercises such as press-ups and burpees. These exercises are rotated through with each exercise performed for 60 s in 3 × 10 min training blocks with 3 min between blocks.

Game-Based Training

Often coaches will give scenario-based instructions to practice particular components of a game. However, in our experience, it is hard to replicate match intensities using open-wicket simulation sessions for all positions. Previously we have recommended that conditioning drills are interspersed within the simulation drills to exceed the match demands. Players can then execute their skills under a higher level of physical demand than actually is required in a game. Researcher Will Vickery has coined the term *battlezone* to describe using an open-wicket training practice with a modified smaller field and with fewer fielders, which is similar to the small-sided games approach used in other sports. This game-based training helps to provide a similar physical and technical match-specific training response (26). Through experimentation it has been found that the physical demands of cricket-specific training can be increased via rule variations, including hit-and-run activities, more so than field size or player number (25).

Manipulations of Interval Training Variables

The running intensity and modality of each HIIT format is systematically adjusted to reach the acute metabolic or locomotor response desired. Examples of adjustment methods include adjusting speeds and distances for straight-line intervals using GPS zones, MAS data, bowler run-up speeds, or increasing sprint zone speeds. For COD intervals, using shuttles with repeat runs between wickets on set times (using testing data and/or MAS calculations) is a commonly employed manipulation method.

Surveillance and Monitoring Tools

The monitoring tools that we've used in cricket are generally limited to player-friendly measures with comparisons made to historical data. Measures include bowling loads, time on feet or session durations, and sessional RPE responses, while neuromuscular-related fatigue is often assessed with countermovement jump protocols and gym-based power monitoring. GPS (distances and speeds) and accelerometry-acquired data have emerged from the research domain and are now used extensively across

the majority of first-class and international teams in the major cricket-playing countries. While logistically challenging in many instances, the use of real-time data capture and in-session feedback is increasing. This is beginning to enhance the quality and accuracy of interval training methods, particularly in the skill domain where the interval training variables are less easily standardized. Finally, we often use the individual responses compared to an individual's history to tailor a player's subsequent training with top-up HIIT training in the case of insufficient fitness.

Strategies for Structuring the Training Program Using HIIT

Strategies for structuring HIIT programming in cricket may be confined by various controllable and uncontrollable factors before progression and periodization can be implemented. This next section discusses some of the nuances of training program and HIIT structuring in cricket.

Controllable and Uncontrollable Factors

The real uncontrollable factor within the training program design is the actual physical stimuli of the match based on the influence of performance to the magnitude of load. As an example, a batter making 100 runs in an ODI match has a very large load compared to the player who is dismissed early in a test match and doesn't do anything for 3 d. Other factors important in programming decisions include the match schedule, traveling requirements, and match locations. As highlighted previously, the congested playing schedule with three different game formats, in addition to the fact that players may be playing for several different teams within a calendar year, makes it challenging to program and keep tabs on what players are doing, especially when they are involved with different professional teams around the globe and may or may not be playing different games. Preseason periods are often completed with smaller squad numbers and with other players joining later for a reduced preseason due to their other playing commitments. Another factor to consider when programming sessions is that elite cricket squads often display a large age range, with some players breaking into the squad in their late teens, whereas others are still playing the game professionally into their late thirties and even early forties.

During the in-season period, training programming needs to be flexible, making the most of opportunities when presented (rain days, early finishes) and adjusted based on players' prior match workloads, while factoring in anticipated upcoming game involvement based on the likely squad rotations. Communication between the medical, conditioning, and coaching staff is crucial to the success of any training program, especially during the in-season period. Being a largely skill-based game, the time allocated to HIIT conditioning will always be much lower in priority than obtaining the required volume of skill-based training. Running-based SIT may be incorporated into an extended warm-up to target speed qualities and provide a stimulus for prevention of hamstring injuries at high speeds. HIIT sessions are sometimes incorporated into the skilled sessions in order to increase the physiological loading and allow players to perform skills with more challenge. An example would be to use a run-based HIIT between two skill sequences, and often this is used with wicketkeepers (HIIT type 3).

Program Structure and Progression

In cricket, there is actually very little periodization in terms of the training cycles, except during the preseason. In-season, the preparation-match-recovery cycles are dictated by the match schedule, however, even players experiencing the same matches will experience very different match involvement and physical loading. The periodization model is more condensed, where the training days between matches are often manipulated and adjusted with the short-, medium-, and longer-term objectives in mind. For these reasons, training prescription needs to be highly flexible and individualized, and to do so requires a sound monitoring program, using the principles outlined in chapters 8 and 9.

Incorporation of HIIT

In the earlier preseason program, strengthening and conditioning often take precedence over the technical training, whereas in the later part of the preseason, the technical and skill-based sessions are prioritized. In the later preseason, HIIT is occasionally combined and incorporated into game-based training and open-wicket practices, whereas in the earlier preseason, HIIT sessions are often programmed as stand-alone conditioning sessions. HIIT sessions,

particularly RST, are gradually progressed to prepare players for the expected playing demands in the early season (bowling overs, number of runs between the wickets). Once into the in-season (which is most of the training year), HIIT prescription is less team based and more individualized.

In-season, when match schedules dictate the training program, there are likely to be numerous combinations of match days within a week ranging from 1 to 6 d. The amount of weekly HIIT prescribed is therefore player dependent and takes into account individual overall playing time and loading history over the recent past and expected short-term future volume. The HIIT prescription is highly individualized and can be achieved through a combination of different HIIT types and formats. Batters are extremely challenging to program for due to the

unknown factor of when and for how long they will bat, and a conservative approach is nearly always used. It is somewhat easier to predict bowlers' expected future bowling loads; however, again, it is unknown how long a player will be required to be in the field (can be multiple days). Conditioning coaches will therefore often prescribe extra strength and HIIT sessions for the 12th man (reserve fielder) on match days.

Sample Training Programs

Figures 24.7 to 24.9 provide examples of HIIT programming during different phases of training, i.e., preseason and in-season. Figures 24.4 to 24.6 show types of HIIT drills incorporated into fielding sessions.

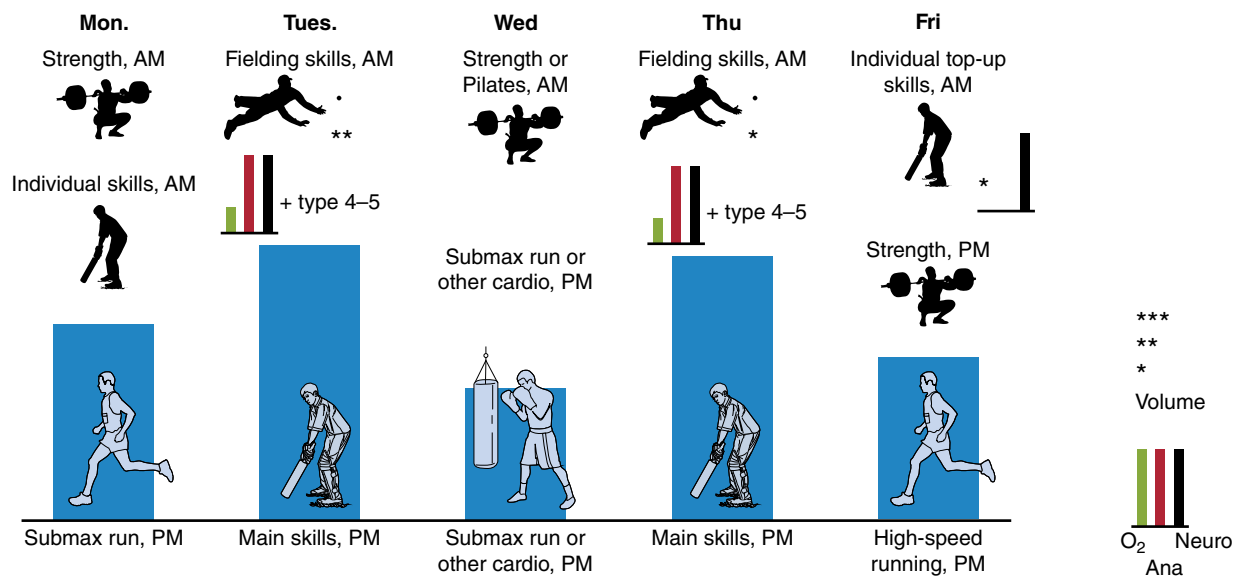


Figure 24.7 Example of preseason programming in an elite team. The blue bars refer to daily training volume (includes technical and tactical in different forms). High-speed running consists of maximal sprint work. HIIT RST is incorporated within fielding drills.

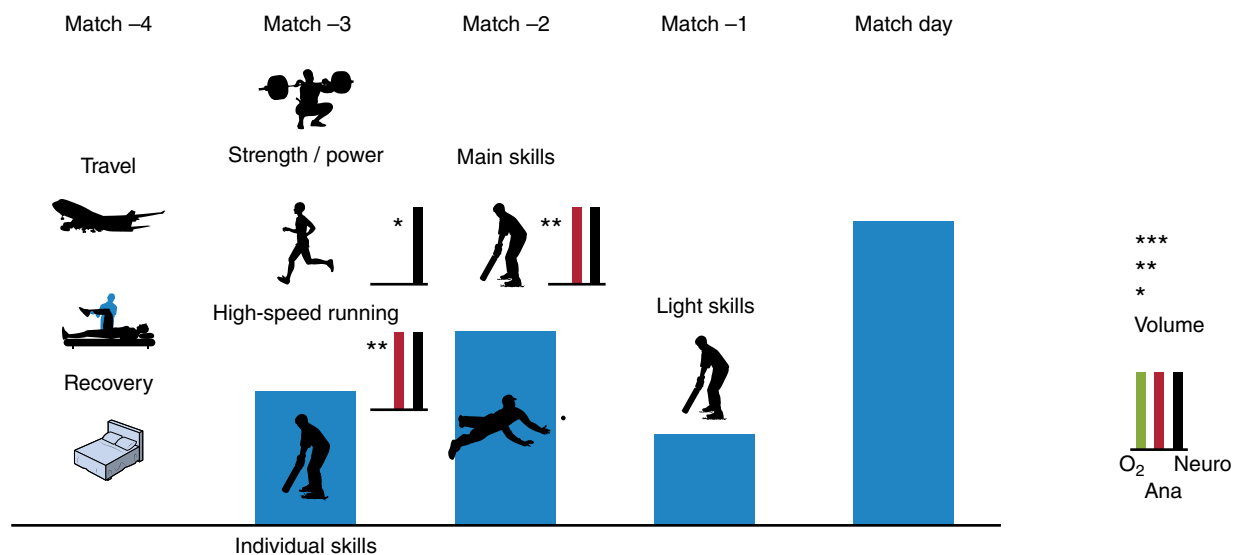


Figure 24.8 Example of in-season programming with one match day for an international team. Training times at international level are determined by home cricket boards, whereby one team will have facility access in the a.m. and the other in the p.m. This will generally be geared toward the home team enjoying the best training times for key days. Interval training is aligned with skill sessions, whereby the timing of strength and recovery modalities will take into account the strength facility access and availability. The blue bars refer to all technical and tactical training volume in different forms. High-speed running consists of maximal sprint work.

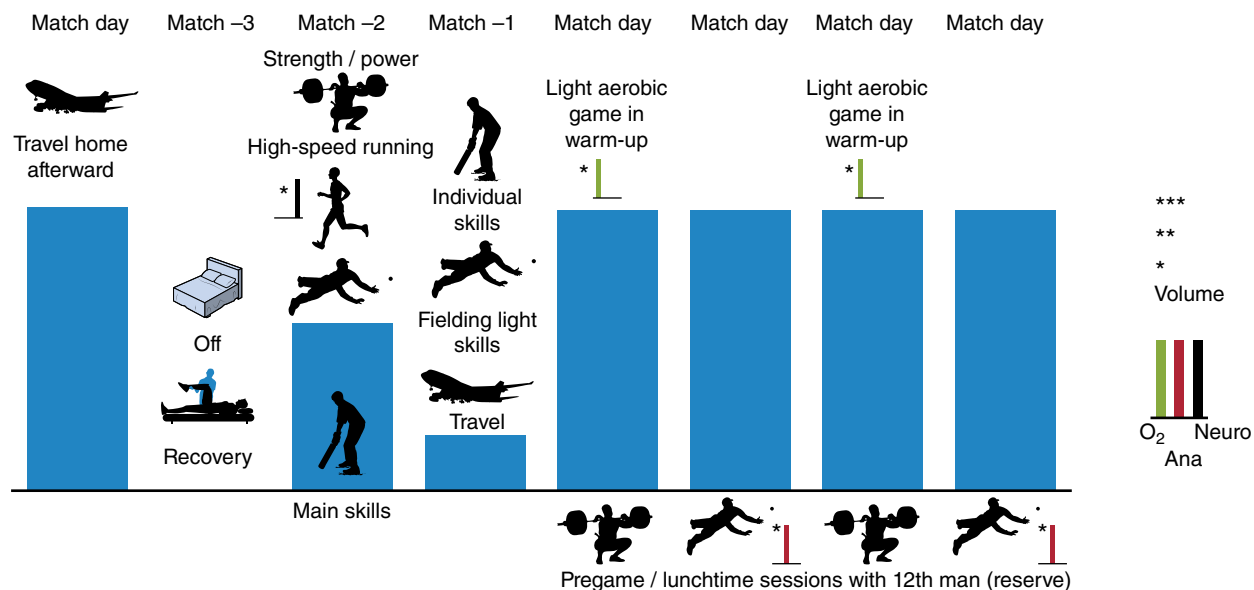


Figure 24.9 Example of in-season programming for a 4 d match for a domestic team. Note, in reality scheduling is not consistent with random numbers of days between matches. The 12th man is decided on the first match day and may or may not stay for the whole 4 d game, depending on where and when the second X1 team is playing. Travel time will vary from 1 to 6 h.

CASE STUDY

The game location and year of occurrence for the following story won't be revealed in the name of confidentiality, but it did occur. Postseason, it is common in a number of countries including Australia, England, and New Zealand for the domestic and national conditioning coaches to get together with the team physiotherapists to discuss best practice, check the annual injury audit data, and hear from invited international experts. In one of these settings, I attended a presentation from Dr. Raphael Brandon, who had recently written his thesis (1) on the neuromuscular response to elite training. The presentation included data and suggestions for the prevention of lower-limb injuries, especially the hamstrings, with the primary recommendation being that in addition to gym-based strengthening, athletes needed to regularly sprint to reduce injury risk.

Injury prevention is a key aspect of any cricket program. Having any player sitting on the sideline injured reduces the overall squad strength and impacts the chance of team success, especially in smaller squads. So in building the annual plan for the upcoming year, we took on-board the recommendations and dutifully prioritized RST both throughout the preseason and for in-season phases. The athletes were easily sold on the idea, and everyone enjoyed the quick sharp sprint session (5 to 8 sprints of 40 to 70 m [44 to 77 yd], with intensity at 90% to 100% with slow walk-back recoveries), which were incorporated into the program at least once a week. However, one particular player was very reluctant to push himself to the required sprinting intensity and would clearly hold back, running at around 70% intensity. Through discussions with the player it became clear that he held a psychological fear about a previous injury from a few years earlier, which he believed might recur if he sprinted. Despite being informed of the rationale for the exercise and consulting with the physiotherapist who also informed him that it was in his best interest to sprint, the player decided that he would prefer to take the chance and risk the consequences rather than push himself to the required 90% to 100% sprint intensities requested to build resiliency. As the player predominantly fielded stationary catching positions, he only infrequently needed to sprint in a game. Privately, the physiotherapist and I considered the athlete's injury risk to be a ticking time bomb.

It was only a few months later, when unfortunately, the athlete's gamble did not pay off. As it happens in cricket, the great unknown for any batter is when he will need to bat, which often can be a substantial time after the warm-up. For this reason, players will often use ladders or perform other activities to keep the body warm and ready while waiting to bat. The other confounding factor is that even during summer, the weather can often be cold, damp, and at times miserable, which it was on this occasion. The player was waiting for a relatively extended time to bat before having to appear in the crease. He started at the nonstriker's end, and literally the first ball bowled was hit, and he was called to sprint through. With the fielder attempting to throw the stumps down at the wicketkeeper's end, our player had to sprint, stretch to slide his bat into the crease, and . . . bang—injured. Of course, we will never actually know if the requested RST conditioning done at the proper intensities would have prevented the injury, but anecdotally speaking, as a squad that year, we had reduced lower-limb injuries in those who followed the prescription properly, and my feeling is that the RST conditioning was an important component of that result.